

Citation-Network Cohesion and Reciprocal Citation Cliques Across Journal-Impact Groups: A Matched Author Analysis

Panagiotis-Alexios Spanakis, Grigorios Alexandrou, and Diomidis Spinellis

Department of Management Science and Technology, Athens University of
Economics and Business (AUEB), Athens, Greece

Abstract

This exploratory study examines how low-impact journals, defined through subject-normalized Eigenfactor percentiles, are associated with denser and more reciprocating patterns of author-to-author citations. Using Crossref records, we assign journals to broad subject areas, compute subject-specific Eigenfactor scores, propagate venue quality to works and authors, match authors in low- (Case) versus high-influence (Control) venues by subject and h_5 , and analyze citation edges for cohesion and anomalies. Across a 10% sample of 9 431 matched pairs, authors in low-impact venues exhibit significantly higher cohesion: $6.7\times$ higher co-author citation rates and $4.7\times$ higher reciprocity in the aggregate Case–Control comparison. A subject-aware hybrid detection pipeline flags 277 outliers with 93.5% Case purity; these outliers display an $11\times$ clique-strength lift relative to non-outliers, revealing a stark “Two Worlds” segregation ($r = 0.71$) where low-impact venues operate as closed citation economies. The largest detected component ($n = 23$) displays a hub-and-spoke topology in which peripheral “Sycophants” funnel citations to central “Beneficiaries” through coordinated bursts, confirming a directed flow imbalance rather than reciprocal exchange among equals. Overall, cohesion, rather than broad asymmetry, accounts for the main Case–Control differences, suggesting that low-impact venues foster segregated, inward-looking

5

10

citation economies that distort bibliometric indicators We ask whether authors in lower-impact journal groups have different citation patterns from matched authors in higher-impact groups. We call the lower-impact observations Cases and the higher-impact observations Controls. We compare 102,626 author–subject pairs within subject and at similar recent citation productivity. All four primary mean differences favored Cases; all four comparisons were significant after adjustment, and 3 median differences were zero. Primary-rule reciprocal-clique membership was 10.96% for Cases and 3.82% for Controls (paired difference 7.14 percentage points; exact $p < 0.001$). Lower-impact-group authors were more likely than matched higher-impact-group authors to belong to formally defined reciprocal citation cliques.

Keywords: scientometrics; low-impact journals; citations; cliques; social network analysis; eigenfactor citation networks; matched design; network cohesion; journal impact; anomaly screening

1 Introduction

Quantitative indicators such as citation counts, the h -index, and journal metrics have become central to the assessment of scholarly impact (Garfield, 2006; Hicks et al., 2015; Hirsch, 2005). These measures offer convenient proxies for influence, but they also create incentives for strategic behavior and manipulation (Baccini et al., 2019; Biagioli & Lippman, 2020; Goodhart, 1984). When deployed uncritically, citation-based indicators risk rewarding volume or visibility over substance, with unintended consequences for research integrity and evaluation systems.

Beyond well-known practices such as excessive self-citation (Van Noorden & Chawla, 2019; Wilhite & Fong, 2012), more coordinated behaviors have emerged. Authors may form tightly connected *citation cliques* — groups that disproportionately reference one another, often within low-impact journals as defined by metrics such as the Journal Impact Factor (Garfield, 1972) or subject-normalized Eigenfactor percentiles (Bergstrom et al., 2008). Such practices distort indicators, undermine trust in the scholarly record, and complicate fair comparison across fields. These dynamics exemplify what has recently been described as *citation orchestration* (Evdaïmon, Ioannidis, et al., 2024), where citations serve strategic rather than intellectual functions.

Citation indicators are widely used to summarize scholarly influence, allocate

resources, and support evaluation. Their convenience also creates incentives to optimize what is measured, which can weaken the connection between an indicator and the quality it is intended to represent (Biagioli et al., 2019; Goodhart, 1984; Hicks et al., 2015). Excessive self-citation, coercive citation, and concentrated citation exchange are therefore important objects of measurement in their own right (Biagioli & Lippman, 2020; Van Noorden & Chawla, 2019; Wilhite & Fong, 2012).

Low-impact journals provide fertile ground for these dynamics. Their emphasis on publication volume, rapid turnaround, and relaxed editorial oversight (Abalkina et al., 2025; Beall, 2012) creates environments where abnormal citation patterns can thrive. Systemic distortions such as national *home bias* (Qiu et al., 2025) further interact with venue incentives, amplifying the risk that rankings and impact assessments reflect coordination rather than genuine uptake of ideas. Most evidence on unusual citation structure is either journal-level or begins with groups already selected for unusual behavior. Journal-network studies can identify exceptional exchange patterns (Kojaku et al., 2021), and analyses of citation stacking show how journal-level citation flows can be distorted (Heneberg, 2016). Author-level concentration indicators can show when a small set of collaborators accounts for much of an author’s citation impact (Evdaimon, Ioannidis, et al., 2024). Less is known about whether simple author-network measures differ across journal-impact groups after authors are compared within subject and at similar levels of recent citation productivity.

This exploratory study investigates whether low-impact journals are significantly associated with cliques and abnormal citation patterns. Using large-scale bibliographic data, we implement a subject-aware pipeline that links journal influence to authors, matches researchers publishing in Bottom- and Top-tier impact journals of comparable productivity (by h_5 index), and analyzes their citation networks for cohesion and anomalies. This study addresses that gap with a matched author–subject design. Low-impact and high-impact refer only to groups defined by subject-specific, publication-weighted 25th and 75th percentile cutoffs of journal Eigenfactor scores. We use *Case* and *Control* as compact labels for

those operational groups; neither label evaluates the scientific merit of an author or work.

We find that extreme anomalies cluster predominantly among authors publishing mostly in low-impact venues: they concentrate in unusually dense, inward-looking structures and form compact outlier groups that set them apart from their Control counterparts. These results provide empirical evidence that venue incentives can shape citation behavior in ways that distort bibliometric indicators. Throughout this paper, authors publishing primarily in low-impact venues are referred to as *Cases*, and their matched counterparts in high-impact venues as *Controls*. Terms such as “syndicate,” “sycophant,” and “beneficiary” are used as *structural labels* describing observed network roles and do not imply intent or misconduct. Similarly we use the terms Control and Case to classify authors publishing mainly in high- or low-impact journals, again without implying any judgement or view regarding the authors’ and their works’ scientific worth. The study asks three questions:

1. Do matched Case and Control authors differ in citation-network cohesion, outgoing citation concentration, and reciprocal-clique membership?
2. Are author–subject profiles that are unusual relative to Controls enriched among Cases?
3. How do fractional citation weights mix across the two groups?

2 Related Work

Prior research has highlighted the fragility of citation-based indicators when used as proxies for scholarly quality. Performance measures not only shape incentives (Goodhart, 1984), but also invite gaming, from excessive self-citation to coordinated cartels (Biagioli & Lippman, 2020; Biagioli et al., 2019). The wider literature on the “metrics culture” documents how quantitative indicators reshape scholarly communication, creating feedback loops between evaluation systems and researcher behavior (Biagioli & Lippman, 2020; Hicks et al., 2015). These mechanisms underpin the emergence of citation cartels (Catanzaro, 2024; Fister Jr. et al., 2016), inflated impact factors (Catanzaro, 2024; Ioannidis et al., 2024), paper mills (Abalkina et al., 2025; Christopher, 2021), AI-generated papers (Spinellis, 2025), and other forms of metric manipulation that erode the interpretive value of bibliometric indicators.

Venue-level frameworks were developed to mitigate some of these distortions. The Eigenfactor metric (Bergstrom et al., 2008) introduced a network-based approach that discounts journal self-citations and weights each citation by the prestige of the citing venue. This recursive weighting sought to capture influence rather than popularity and to counteract local citation loops. However, later analyses have shown that such corrections remain sensitive to database coverage, field size, and reference density (Baccini et al., 2019), indicating that structural biases persist even in sophisticated venue-level measures. We expect Case author-subjects to have higher values than Controls on four primary measures: citations to previous collaborators, reciprocal citation, local neighbor connections, and concentration of outgoing citations. This is an associational hypothesis, not a causal or integrity claim. Because many values are zero, a zero median shift does not rule out a directional difference. Network closure can reflect legitimate specialization and collaboration, so the anomaly procedure is a screening device.

At the author level, organized citation exchanges have been formalized as *citation orchestration* (Evdaimon, Ioannidis, et al., 2024). This framework quantifies concentrated citation flows through indicators such as the ratio of total citations to the squared author's

2 Related Work

Research evaluation based on citation counts, the h -index(C/h^2), the number of distinct authors accounting for half of one's citations ($A50\%C$), and the count of hyper-collaborators sharing over fifty joint papers ($A50$). Together, these metrics reveal unusually cohesive collaboration networks that may signal collective metric inflation. Earlier studies of citation cartels and excessive self-referencing (Baccini et al., 2019; Fister Jr. et al., 2016; Heneberg, 2016; Ioannidis, 2025; Kojaku et al., 2021) align with this view, emphasizing how reciprocal ties among small groups of authors can yield disproportionate bibliometric visibility. Such coordinated behaviors blur the line between collaboration and manipulation, challenging the assumption that citation networks represent organic intellectual influence, and journal indicators has long been accompanied by cautions about field differences and strategic responses (Garfield, 2006; Hicks et al., 2015; Hirsch, 2005). Self-citation is common and can be substantively appropriate, but extreme rates complicate

comparisons (Aksnes, 2003; Van Noorden & Chawla, 2019). Coercive citation and metric gaming demonstrate that observed references can also respond to editorial and institutional incentives (Baccini et al., 2019; Wilhite & Fong, 2012).

Systemic and infrastructural distortions further complicate these dynamics. National *home bias* inflates domestic citation counts, particularly in large scientific systems such as China’s (Qiu et al., 2025), while technical manipulation has emerged through practices such as “sneaked references”, which are inserted into metadata but absent from article text (Besançon, Cabanac, et al., 2024). Industrial-scale paper mills produce fabricated papers that populate citation networks with artificial linkages (Abalkina et al., 2025). Recent investigations reveal even more elaborate schemes, including the creation of fictitious author identities and peer reviewers to infiltrate legitimate journals. As reported by Naddaf (2025), these fake personas — sometimes maintained across multiple journals — enable mills to generate both papers and favorable reviews, prompting publishers to explore researcher identity verification frameworks and persistent identifier checks such as ORCID trust markers.

The present research extends this discourse by systematically examining the relationship between author-level citation behaviors and the influence level of the publication venue. Several strands of work motivate a network

perspective. Kojaku et al. identify anomalous groups in journal citation networks by comparing observed patterns with network expectations (Kojaku et al., 2021). Heneberg examines the transition from journal self-citation to citation stacking through journal-level citation flows (Heneberg, 2016). These approaches establish the value of relational structure but operate at the venue level. At the author level, Evdaimon et al. propose indicators based on the concentration of citing authors and hyper-collaboration (Evdaimon, Ioannidis, et al., 2024). Their concentration measures identify a narrow citing base; local closure, while controlling

for individual productivity (h_5 index) and disciplinary norms through a matched Case–Control design. While previous studies have primarily investigated the orchestration of specific bibliometric indicators (Evdaimon, Ioannidis, et al., 2024) or semantic irregularities within article metadata (J. Liu et al., 2024), this study identifies *structural cohesion*, which manifests as dense local clustering and triadic closure, as the primary diagnostic signature of citation anomalies within low-impact journals. By implementing a subject-aware isolation forest on author-to-author networks across a 10% sample of 9 431 matched pairs, the analysis demonstrates that such cliques constitute structural byproducts of venue-level incentives. Consequently, this approach

provides a network-centric diagnostic framework for identifying organized citation manipulation practices that evade traditional bibliometric oversight by focusing on network closure patterns rather than simpler citation calculations. reciprocity, and 145 concentration of an author’s own references provide complementary information about how that base is connected.

3 Methodology

The present analysis combines this network perspective with within-subject author matching. It does not attempt to classify behavior as legitimate or 150 illegitimate. Instead, it estimates matched differences in a small, declared set of network measures and applies a Control-referenced screen whose output remains explicitly descriptive.

We developed a subject-area-aware pipeline linking journal quality to author citation behavior . The main steps were: (i) journal subject classification, (ii) computation of subject-specific journal Eigenfactor scores, (iii) propagation of venue impact 155 to works and authors, (iv) matched pairing of Case and Control authors based on journal impact and productivity, and (v) construction of author–author citation networks with cohesion and anomaly features. Figure ?? illustrates the main analysis subgraph of this pipeline.

3 Data and Cohort

To ensure consistency, replicability, and correctness across all stages of the data pipeline before statistical analysis, these were 160 orchestrated through *simple-rolap* (Spinellis, 2024b) and unit-tested through *RDBUnit* (Spinellis, 2024a).

3.1 Source data and study window

Analysis DAG for the social-network stage. Ellipses represent derived tables and arrows indicate dependencies. Curated inputs feed `citation_network_final` and downstream aggregates.

3.2 Data and Preprocessing

Data collection. Crossref metadata were ingested via *alexandria3k* (Spinellis, 2023), covering works published between 2020 and 2024. We retained DOIs, years, titles, abstracts, references, author identifiers (ORCID), and normalized ISSN for journals. Duplicates and inconsistencies were cleaned, and derived layers linked works to authors, journals, and citations. Appendix ?? documents the full data derivation pipeline and the filtering decisions applied at each stage.

The source is a fixed Crossref snapshot processed with *alexandria3k* (Spinellis, 2023). The study window covers citing and cited works dated 2020–2024. Available fields include Digital Object Identifiers (DOIs), publication year, normalized journal International Standard Serial Numbers (ISSNs), work references, and author rows with optional ORCID researcher identifiers. ORCID is a persistent identifier for a researcher. The estimates apply to this fixed snapshot and to the author–subject observations that meet the stated inclusion criteria.

Journal Subject Classification. Each journal was assigned to one of five broad subject areas: 1) Journals are assigned to Health Sciences, 2) Life Sciences, 3) Physical Sciences, 4) Social Sciences & Humanities, and 5) Multidisciplinary, using an LLM as an annotator (LLM Guidelines Consortium, 2024). This taxonomy mirrors the broad subject areas of the All Science Journal Classification (ASJC) scheme used by Scopus (Burnham, 2006), one of the most widely adopted classification frameworks in scientometrics. Separating subject areas is essential for computing meaningful within-field Eigenfactor percentiles, as citation practices vary substantially across disciplines (Bergstrom et al., 2008).

Classification was performed using OpenAI gpt-4.1 (snapshot gpt-4.1-2025-04-14, accessed June 2025) with zero temperature and few-shot exemplars. The model processed journals in batches of ten ISSNs per API call, supplying for each journal up to five representative works ordered by citation count, so that the most influential works appeared first. Prompts and decision rules prioritized abstracts over titles and journal names; when an abstract was unavailable, classification relied on the journal name and title alone. To ensure reproducibility, we used a fixed random number generation seed and cached predictions to JSON files. The full prompt template and representative input–output pairs are included in the replication package (Spanakis et al., 2026). No open-weight baseline model was evaluated; this is acknowledged as a limitation (Section 10).

Social Sciences and Humanities, or Multidisciplinary. The existing classification uses a deterministic, cached large-language-model annotation of representative journal content.

Eigenfactor Computation. To quantify journal impact, we computed The retained venue classification uses fixed subject-specific Eigenfactor scores (Bergstrom et al., 2008) on journal citation networks (journal scores from a separate directed journal-citation calculation (Bergstrom et al., 2008). An Eigenfactor score is a journal influence score based on citation flows.195 In plain terms, the calculation passes influence from citing journals to cited journals, excludes journal self-citations, and uses article counts to redistribute influence from journals without outgoing citations and to restart the calculation. These journal-level scores define the two venue groups; they are an input to the comparison, not an author-network outcome. The exact matrix specification is 200 given in Appendix A.

The current workflow consumes the fixed database scores as an upstream exposure rather than re-estimating them. After scores are attached to eligible 2020–2024), excluding self-citations. The computation used standard power iteration and produced normalized influence scores work rows, the 25th and 75th percentile cutoffs are calculated across positive-205 score work rows within each subject.

Defining Venue Impact. Within each subject, journals in the top quartile of Eigenfactor were labeled *high-impact*, and those in the bottom quartile *low-impact*. Labels propagate to works (by ISSN) and to authors through their portfolios. Alternative thresholds and persistence checks were evaluated for robustness. A journal therefore contributes once per eligible work: the cutoffs are publication weighted, not percentiles over 210 distinct journals. Journals at or below the lower cutoff form the low-impact group and those at or above the upper cutoff form the high-impact group. No new venue classification was undertaken for this revision.

3.2 Author Matching and Network Construction Author–subject cohort

Author Tiering and Matching. Authors were classified as **Control** if most of their output was in high-impact journals, **Case** 215 if concentrated The unit used throughout is an *author–subject*, meaning one researcher considered within one subject area, identified by (ORCID, s). An author is a

Case in subject s when at least 70% of eligible works in that subject appear in low-impact journals, and **Mixed** otherwise. Mixed-tier authors were excluded from the matched-pair analysis to ensure clean separation between venue quality levels. We then matched Case authors to Control authors of similar productivity (subject-specific h_5 index), ensuring comparability in output volume and field. **a Control when at least 70% appear in high-impact journals. Authors not meeting either rule are excluded from matching. A uniqueness check prevents more than one group assignment for the same (ORCID, s) key.**

Sample and Availability. The final analytical dataset comprises a 10% sample of 9 431 **The primary cohort contains 102,626 matched Case–Control author pairs** (see Table ?? for the per-subject breakdown). All pair-level and network-level aggregates were computed per subject area and combined only after validation of within-subject consistency. All processed data tables are available through the public replication repository described in Section ?. **pairs across the five subjects. It is built from every non-empty ORCID in the fixed snapshot that meets the portfolio, work-count, and positive- h_5 eligibility rules; no final-character sampling rule is applied.**

Citation Network and Metrics. For matched authors, we constructed directed author-to-author citation networks (

4 Matching

Recent subject-specific citation productivity is summarized by h_5 , the Hirsch index calculated from an author’s works in subject s published during 2020–2024).

Each citation edge between a citing work with n_c authors and a cited work with n_a authors received weight $w = 1/(n_c \times n_a)$, correcting for team size. From these edges we derived cohesion metrics (self-citation, co-author citation, clustering, triangles, k -core, eigenvector centrality) and pairwise anomaly metrics (reciprocity, asymmetry, velocity, citation bursts). These measures capture both dense reciprocal structures and asymmetric one-way flows. During exploratory analysis, raw pairwise metrics (maximum asymmetry, average velocity, and maximum burst) were also computed; in the formal pipeline they were replaced by normalised variants that are comparable across activity levels: *citation balance* captures net giver/receiver bias as a bounded $(-1, +1)$ ratio, while *normalised burst intensity* adjusts the largest single-peer citation weight by total incoming volume. Table ??

(Appendix ??) lists the 14 features employed for statistical testing and modelling. (Hirsch, 2005). In ordinary language, h_5 is the largest number of works that each have at least that many citations in the fixed snapshot. The matching pool requires $h_5 \geq 1$ and excludes 245 otherwise group-eligible zero- h_5 profiles before candidate generation.

4.1 Statistical Analysis and Outlier Detection

We pair lower-impact and higher-impact author–subjects within the same subject when their h_5 values differ by at most three points. Candidates are considered in the fixed order subject, smallest absolute h_5 difference, Case ORCID, and 250 Control ORCID; each author–subject can be used only once.

Statistical Analysis and Robustness. Matched pairs were compared using Wilcoxon signed-rank tests (Wilcoxon, 1945), with Benjamini-Hochberg FDR correction (Benjamini & Hochberg, 1995). Effect sizes were quantified via Cliff’s δ (Cliff, 1993) and Cohen’s d (Cohen, 1988), with confidence intervals estimated through bootstrap resampling (Efron & Tibshirani, 1993). Robustness checks included permutation tests, placebo pairing, random-effects meta-analysis across subjects, and 255 alternative outlier detectors (Isolation Forest (F. T. Liu et al., 2008), Local Outlier Factor (Breunig et al., 2000), One-Class SVM). Sensitivity analysis utilized z-score thresholds ($1\sigma, 2\sigma, 3\sigma, 4\sigma$) relative to the control baseline to establish interpretable detection cutoffs. Community detection used the Louvain method (Blondel et al., 2008), which optimises the modularity criterion (Newman & Girvan, 2004). An independent audit repeats the deterministic match from the generated full-cohort profiles and compares it with the materialized 260 pairs. It requires identical order and membership, no pair beyond the three-point limit, and no reused author–subject key. The complete audit accompanies the replication materials.

The matching table supplies the Case ORCID, Control ORCID, subject, and both h_5 values. Feature rows are linked back to this table twice, on $(\text{case_orcid}, s)$ 265 and $(\text{control_orcid}, s)$; ORCID-only joins are not used.

Outlier Detection Procedure. We developed a subject-aware hybrid outlier detection strategy linking multivariate anomalies to specific cohesion signatures. The pipeline combines an Isolation Forest (IF) (F. T. Liu et al., 2008) with a domain-specific

Table 1: Matched authors by subject.

Subject	Pairs	h_5 observed	Exact h_5	Median Δh_5	Median $ \Delta h_5 $	Maximum $ \Delta h_5 $
Overall	102,626	102,626	64,176	0.00	0.00	2
1	40,439	40,439	17,869	-1.00	1.00	2
2	28,045	28,045	15,205	0.00	0.00	2
3	3,383	3,383	3,345	0.00	0.00	1
4	24,879	24,879	21,877	0.00	0.00	1
5	5,880	5,880	5,880	0.00	0.00	0

Δh_5 is Case minus Control. Every pair must satisfy $|\Delta h_5| \leq 3$. Exact- h_5 pairs form a sensitivity cohort; they do not replace the primary matched cohort.

Cohesion Composite Score. The IF model used standard hyperparameters (`n_estimators=200`, `contamination=0.01`) to capture

270 general anomalies. The Cohesion Score was computed as the standardized sum of We report the distribution of h_5 and the within-pair difference before outcome testing. Because residual differences can remain under caliper matching, the four most discriminative features: co-author citation rate, clique strength, reciprocity, and outgoing HHI¹ four primary comparisons are repeated on the subset with exact h_5 equality.

275 We computed 14 behavioural features per author (Table ??). Of these, normalised burst intensity was available for only 7711 of

The generated cohort audit found a maximum observed absolute h_5 difference of 2 and 0 pairs beyond the three-point caliper.

5 Network Construction and Definitions

280 We represent authors as nodes and citations as links between nodes. A link has a direction, from the citing author to the 9431 matched pairs (18% missing) because many authors recorded no burst events in the study window. Statistical tests handle this by dropping incomplete pairs per metric; however, the Isolation Forest requires a complete feature matrix, and zero-imputing 18% of values would inject a spurious signal. The model was therefore trained on the remaining 13 features. All features were standardised (zero mean, unit variance) and multiplied

285 by hand-tuned weights reflecting Random-Forest importance and Wilcoxon effect sizes (e.g., Co-author Rate weighted 4.0); authority metrics were sign-inverted so that lower values indicate greater suspicion. An author was flagged as an outlier only

¹The Herfindahl-Hirschman Index (HHI) measures concentration. In this context, it quantifies the extent to which an author repeatedly cites the same narrow set of recipients, indicating a lack of source diversity.

if they satisfied both conditions: an IF anomaly score above the threshold **AND** a Cohesion z-score $> 4\sigma$ (relative to the Control baseline). Sensitivity tests evaluated thresholds from 1σ to 4σ to confirm the robustness of detected syndicates. Outlier labels were merged back into the unified author-feature table for downstream network analyses (results in Section ??). **cited author**, and a fractional weight that accounts for the number of authors on both 290 works. We use the word “network” for this set of nodes and weighted links.

6 Findings

5.1 Reference resolution and fractional weights

This section presents the main results of the matched Case–Control analysis. We first examine overall and field-specific patterns of cohesion and asymmetry, then derive unsupervised behavioral archetypes and identify statistical outliers using a subject-aware 295 Isolation Forest. Together, these analyses show that cohesion—rather than sustained one-sided citation flows—is the primary structural distinction between authors publishing in low- and high-influence venues. The main network features used were the *clustering coefficient* (density of local interconnections); *triangles* (closed three-way citation structures); the *k-core number* (depth of embedding in dense subnetworks); *betweenness centrality* (how often an author lies on shortest paths, a proxy for brokerage **A** *citing-work*/coordination potential); *velocity* (rate of citation accumulation); *burst* (sudden within-window surges of 300 citations); **cited-work DOI pair** is deduplicated before either work is expanded to author rows. Let n_c and *asymmetry* (imbalance in giving vs. receiving citations). n_a be the complete counts of author rows on the citing and cited work. These denominators include authors without ORCIDs. A resolved work citation contributes

$$\frac{1}{n_c n_a} \tag{1}$$

to each identified citing-author/identified cited-author combination. Rows with 305 an unidentified endpoint do not produce analytic author edges; in particular, missing identifiers cannot become a shared pseudo-author. Multiple matched authors on one citing work do not multiply the underlying resolved work reference.

5.2 All Subjects Combined

Every analytic edge is keyed by subject, citing ORCID, cited ORCID, and citation year. Only citing works assigned to the matched author’s subject contribute to that author–subject’s measures. Duplicate rows are aggregated so that this key is unique.

At the aggregate level, authors publishing primarily in low-impact venues display markedly stronger *cohesion*: on average, they cite co-authors **6.7× more frequently** ($p = 9.1 \times 10^{-66}$), exhibit **4.7× higher reciprocity rates** ($p = 2.7 \times 10^{-44}$), For coauthor classification, let c_{ij} be the first year in which authors i and show **3.1× greater outgoing concentration** (tunnel vision, $p \approx 0$). Figure ?? displays the mean paired difference in co-author citation rates across subject fields. j published together. A citation from i to j in year t is a prior-coauthor citation only if $c_{ij} < t$. A sensitivity analysis uses $c_{ij} \leq t$ to show the effect of admitting same-year collaborations.

Co-author citation gap by subject field. Forest plot of mean paired difference $\Delta = \bar{r}_{\text{Case}} - \bar{r}_{\text{Control}}$ in co-author citation rate, with 95% confidence intervals. Marker $\bullet$ = field estimate; $\blacklozenge$ = overall. Δ labels report rounded effect sizes. The dashed vertical line marks zero (no difference); the pale red shading covers the positive region where Case authors exceed Controls.

5.2 Graph and reference populations

Table ?? summarises the aggregate Case–Control contrast across all 9431 matched pairs and illustrates it with a representative pair drawn from Health Sciences. The analysis uses three views of the citation links so that each measure has a clear population.

Each author’s outgoing links. For each matched author–subject i , this view contains citations from i to any identified author reached from an eligible citing work. It defines self-citation, citations to previous collaborators, outgoing concentration, and the largest year-to-year increase to one recipient.

Links among matched authors. For each subject s , this view retains only matched

authors and the links between them. It defines reciprocal citation, local neighbor connections, the balance of incoming and outgoing weight, and citation mixing between the two groups.

Deduplicated work references. This view is built before author expansion and defines same-journal referencing even when the cited author is not identified.

Keeping these views separate prevents a measure based on all identified recipients or work references from being mistaken for a measure based only on matched authors.

5.3 Notation, metrics, and missing values

The cohesion gap is not driven by a few extreme outliers but by a broad rightward shift across all four dimensions (co-author citation rate, Some measures follow an author’s outgoing citations to any identified recipient; others use only links between matched authors. Direction is ignored only for the simple neighbor graph used for local clustering.

Let x_{ijt} denote the fractional citation weight from author i to author j in year t , after the construction above. Missing dyad-years are zero, and $w_{ij} = \sum_{t=2020}^{2024} x_{ijt}$ is cumulative weight. If V_s is the set of matched authors in subject s , the matched-author graph is the directed, weighted, loopless graph

$$G_s = (V_s, E_s, w), \quad E_s = \{(i, j) : i, j \in V_s, i \neq j, w_{ij} > 0\}. \quad (2)$$

Its simple binary undirected projection is

$$U_s = (V_s, \tilde{E}_s), \quad \{i, j\} \in \tilde{E}_s \iff i \neq j \text{ and } w_{ij} + w_{ji} > 0. \quad (3)$$

Direction and fractional weight are discarded only in U_s . Self-edges remain available in the outgoing ego layer for self-citation rate, outgoing HHI, and local clustering).

Permutation testing (10 000 label shuffles of Case–Control assignments) independently confirms the effect is far outside the null

355 distribution ($p < 0.001$). and surge share but do not enter G_s or its peer-network measures.

In contrast to their high internal cohesion, Case authors exhibit significantly lower authority metrics. Controls exhibit **13× higher eigenvector centrality** than Cases ($p = 3.4 \times 10^{-132}$), and **2.2× higher Citation Entropy**, indicating that while Cases have a similar citation volume, they lack broad authoritative influence. Note that the example pair in Table ?? both show near-zero eigenvector centrality at

360 5.4 Formal reciprocal-clique analysis

We test citation-clique structure on all matched authors, not only on authors flagged by the screen. A clique C is a group of at least four matched authors in which every pair is connected by a citation in at least one direction. We keep only maximal groups, meaning that no additional matched author can be
365 added without breaking that rule. The direction and weight of the $\times 10^5$ scale, which is consistent with the right-skewed distribution of this metric: the aggregate mean is driven by a small number of highly connected authors, and individual pairs can deviate substantially from the population average. This divergence is further supported by the random forest feature importances in Figure ??, where cohesion and concentration metrics rank as the most discriminative features for tier classification, while authority metrics such as eigenvector centrality play a secondary role. links are then
370 evaluated separately. For a clique of size k , directed density is

$$d_C = \frac{m_C}{k(k-1)}, \quad (4)$$

where m_C is the number of positive directed dyads among the k members.

Weighted reciprocity is

$$r_C = \frac{\sum_{\{i,j\} \subset C} \min(w_{ij}, w_{ji})}{\sum_{\{i,j\} \subset C} \max(w_{ij}, w_{ji})}. \quad (5)$$

The primary rule requires at least **75%** of all possible directed links and a weighted reciprocity of at least **0.50**. We repeat it for minimum clique sizes $k = 3, 4, 5$ and
375 reciprocity thresholds 0.25, 0.50, 0.75; the density threshold remains fixed at **0.75**.

Random Forest feature importances. Horizontal bars show mean decrease in impurity for 13 of the 14 behavioural features listed in Table ??, sorted by importance. Bars are coloured by feature category (see legend). Inline labels give raw importance score and share of total.

The Case–Control gap is explained mainly by distinct behavioral signatures. These patterns persist under conservative matched-pair comparisons: **12 out of 14** tested behavioral metrics remain significant after Benjamini-Hochberg FDR correction ($\alpha = 0.05$) (Benjamini & Hochberg, 1995). This systemic disparity is not static: cumulative citation trajectories (2020–present) show the Case–Control gap *widening* over time, indicating an accelerating divergence. For inference, overlapping qualifying cliques are converted to one binary membership value per author–subject. Case and Control membership rates are compared within the original matched pairs. The exact two-sided null distribution conditions on the number of discordant pairs and assigns either member of each discordant pair equal probability of carrying the clique-membership label. We report both discordant-pair counts, the paired rate difference, the exact p -value, and the direction of the rate difference within each subject.

5.5 Subject-Stratified Effects

390

The study-specific temporal measure is

$$B_i = \frac{\max_{j,t=2021,\dots,2024} [x_{ijt} - x_{ij,t-1}]_+}{\sum_j \sum_{t=2020}^{2024} x_{ijt}}. \quad (6)$$

This study-specific temporal increment is not a state-transition burst-detection model.

Enhanced Case cohesion is a recurrent feature across subjects, but its expression varies by field (Figure ??). In the health and life sciences, the divergence appears mainly as consistently higher co-author citation and stronger local clustering. In multidisciplinary journals and physical sciences, dense networking more often coincides with short, episodic activity waves. In the social sciences and humanities, cohesion signals are subtler, but sharper temporal concentrations of citations within short windows are more prominent.

Table 2: Declared author–subject measures.

Measure	Definition	Population and interpretation
Prior-coauthor citation rate	$\frac{\sum_{t,j \neq i} x_{ijt} \mathbf{1}(c_{ij} < t)}{\sum_{t,j \neq i} x_{ijt}}$	Outgoing ego layer. Fraction of identified non-self outgoing weight directed to an author whose first collaboration preceded the citation year.
Self-citation rate	$\frac{\sum_t x_{iit}}{\sum_{t,j} x_{ijt}}$	Outgoing ego layer. Fraction of identified outgoing weight assigned to the same ORCID.
Reciprocity	$\frac{\sum_{j \neq i} \min(w_{ij}, w_{ji})}{\sum_{j \neq i} w_{ij}}$	Directed, weighted G_s . Among matched peers, the share of outgoing weight that is also returned to the author.
Local clustering	$\frac{2T_i}{k_i(k_i - 1)}$	Binary projection U_s . Among an author’s matched neighbors, the share of neighbor pairs that are connected to each other. Here k_i is the number of matched neighbors and T_i is the number of connected pairs among those neighbors.
Outgoing Herfindahl–Hirschman index (HHI)	$\sum_{j \neq i} p_{ij}^2, p_{ij} = w_{ij} / \sum_{k \neq i} w_{ik}$	Outgoing ego layer. Concentration of identified non-self outgoing weight across recipients; values near 1 mean that one recipient receives nearly all of the weight (Hirschman, 1964).
Journal endogamy	Resolved outgoing work references whose citing and cited journals have the same nonblank normalized ISSN, divided by all resolved outgoing work references	Deduplicated work-reference layer. Share of resolved references whose citing and cited journals have the same normalized ISSN. This descriptive term does not imply impropriety.
Within-sample citation balance	$\frac{\sum_{j \neq i} w_{ij} - \sum_{j \neq i} w_{ji}}{\sum_{j \neq i} w_{ij} + \sum_{j \neq i} w_{ji}}$	Directed, weighted G_s . Positive values mean more outgoing than incoming weight among matched peers; negative values mean more incoming weight.
Maximum annual dyadic surge share	$\frac{\max_{t=2021, \dots, 2024} \sum_j [x_{ijt} - x_{ij,t-1}]_+}{\sum_j \sum_{t=2020}^{2024} x_{ijt}}$	Outgoing ego layer. Largest year-to-year increase to one recipient, divided by all identified outgoing weight.

Notes. x_{ijt} is fractional citation weight, w_{ij} is cumulative weight, c_{ij} is the first joint-publication year, and $\mathbf{1}(\cdot)$ is an indicator. G_s is the directed matched-author graph, U_s is its binary neighbor graph, and $[z]_+ = \max(z, 0)$. Measures are calculated separately for each author–subject; the zero and missing-value rules follow the table.

Unavailable values remain missing through feature assembly. A zero is used only when the denominator and graph population are observed but the relevant 400 event is absent. Thus a defined rate with no qualifying citations is zero; reciprocity is zero when internal outgoing weight exists but none is returned; clustering is zero for an observed induced-graph node with fewer than two neighbors; and journal endogamy is zero when resolved references exist but none remains within the citing journal. Outgoing HHI is missing when there is no identified non-self 405 outgoing weight, balance is missing when total internal incident weight is zero, journal endogamy is missing when no resolved outgoing work reference exists, and surge share is missing when total identified outgoing weight is zero. These rules imply $0 \leq B_i \leq 1$. Statistical comparisons use complete-pair deletion separately for each metric rather than a common imputed matrix. 410

Across fields, one-sided asymmetry measures remain generally subdued. Where differences emerge, they align more with transient speed-ups (velocity/bursts) than with persistent directional imbalance.

Subject-Specific Effect Sizes (Cliff’s δ). Each cell reports the effect size for a given metric within a subject area. Positive values (white–pale red) indicate that Cases exhibit higher values than Controls. Negative values (blue) indicate that Controls exhibit higher values than Cases. Co-author Citation Rate shows near-zero or slightly positive values across all subjects, while velocity and burstiness are uniformly negative, with Life Sciences displaying the largest deficit in burstiness ($\delta = -0.73$). *Note.* While ‘Cohesion’ (Co-author Rate) is universally high, other traits like ‘Burstiness’ vary by field, confirming the ‘Subject-Specific Accents’ hypothesis.

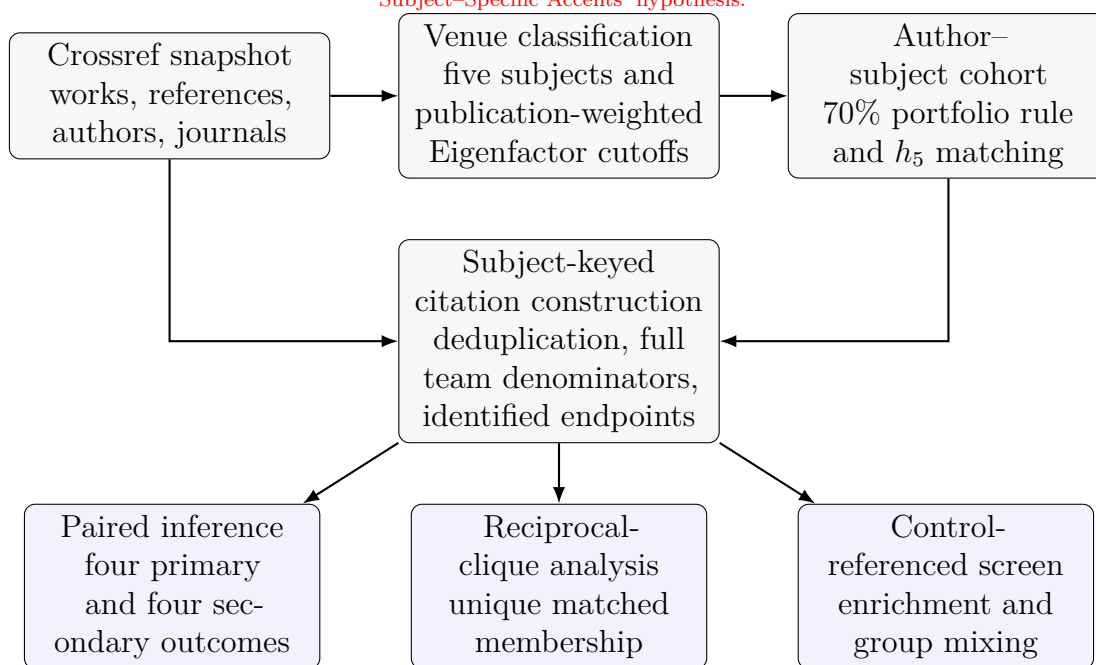


Figure 1: **Reader-facing study design.** Venue classification and author matching define the cohort; subject-keyed citation construction then supports the paired, clique, and screening analyses.

6 Paired Inference

The signature of Case authors is greater network cohesion. Whether this manifests as intensified co-author referencing, increased local triadic closure, or episodic waves is field-dependent.

415

6.1 Behavioral Archetypes

K-means clustering on the standardised behavioural features reveals a clear three-way typology: **Central** (structurally prominent, well-embedded authors), **Independent** (typical, low-cohesion), and **Solo** (extreme self- The four primary outcomes are citations to previous collaborators, reciprocal citation, connections among an author’s citation neighbors, and co-author citation). Both tiers are dominated by the Independent profile, but the two peripheral clusters split along tier lines: Controls are nearly twice as likely to fall in the Central cluster (19.0% vs 10.7%), whereas the Solo cluster is tiny but almost exclusive to Cases (27 of 29 authors). Thus K-means isolates a small, extreme Case-only archetype, while the broader cohesion gap reported in Section ?? manifests as a within-Independent shift rather than a distinct cluster. **outgoing HHI**. HHI is a concentration score from 0 to 1; it is higher when an author sends most outgoing citation weight to a small number of recipients. The four secondary outcomes are self-citation, same-journal referencing, the balance of incoming and outgoing links, and the largest year-to-year increase to one recipient. The former normalised burst-intensity feature is not an additional outcome; it was replaced by the surge share in Equation 6. The revised analysis therefore reports eight distinct outcomes in two declared families.

This divergence is consistent with clique-like behavior: Case authors follow multiple higher-cohesion strategies, whereas Controls converge on a single low-cohesion, outward-looking profile. Figure ?? illustrates this separation: the Linear Discriminant Analysis (LDA) projection reveals minimal overlap between tiers, with Cases clustering toward the Cohesion-driven extreme and Controls toward the Authority-driven regime. For each matched pair, we subtract the Control value from the Case value. A positive difference therefore means that the Case value is larger. Only pairs with both values available enter a comparison. Because

435

the measures contain many ties and zeros, we use the two-sided paired Wilcoxon signed-rank test, which does not assume a normal distribution (Wilcoxon, 1945).

440 We report the number of pairs, the median difference, and a rank effect size:

$$r_{rb} = \frac{W^+ - W^-}{W^+ + W^-}, \quad (7)$$

where W^+ and W^- add the ranks of positive and negative differences after zero differences are omitted. The effect ranges from -1 to 1 ; a positive value means that Case ranks tend to be higher. We obtain 95% intervals from 2,000 bootstrap samples of matched pairs (Efron & Tibshirani, 1993). We also report mean
445 differences and their bootstrap intervals because they give an intuitive measure of the size of the shift. Benjamini–Hochberg correction adjusts the four p -values in each outcome family for the number of tests (Benjamini & Hochberg, 1995).

Tier separability: LDA projection. Kernel density estimates of the scores produced by projecting each author’s standardised feature vector onto the single Linear Discriminant Analysis (LDA) axis that maximally separates Case from
450 Control authors. Dotted vertical lines mark within-tier medians. Negative scores correspond to the cohesion-driven regime (higher co-author citation, clustering, and reciprocity), while positive scores correspond to the independent regime (lower cohesion, typical citing behaviour). As a randomization check, we randomly reverse the sign of each pair difference in 10,000 runs and compare the resulting mean differences with the observed one. This keeps the size of every pair difference while removing its
455 direction. We also repeat the primary comparisons for exact- h_5 matches and for a definition that counts collaborations formed in the citation year.

Three behavioral profiles emerge from the clustering:

1. **Central** ($n = 2\,807$; 10.7% of Cases, 19.0% of Controls): high structural prominence (k-core, triadic closure) combined with pronounced citation asymmetry; more prevalent among Controls, consistent with the embedded network position of
460 top-tier authors.
2. **Independent** ($n = 16\,026$; 89.0% of Cases, 80.9% of Controls): low cohesion and connectivity; the dominant profile for

both tiers.

3. **Solo** ($n = 29$; 0.3 % of Cases, < 0.1 % of Controls): extreme self-citation, co-author citation, and reciprocity; nearly exclusive to Case authors (27 of 29). The cluster is small in absolute terms, so this archetype is descriptive rather than inferential.

465

The Solo pattern—characterised by co-author citation rates nearly 200 times the population mean —appears almost exclusively among Cases. Controls are proportionally more represented in the Central cluster, consistent with the structural prominence of top-tier authors. Both tiers are dominated by the Independent baseline profile.

7 Anomaly Screen

7.1 Outlier Detection Results

470

The anomaly analysis asks a different question from the matched comparison: are unusual profiles more common among Cases than among Controls? It is a screening rule, not a prediction model and not evidence of intent.

Using the hybrid detection strategy at the 4σ threshold, we identified 277 *high-confidence outliers*. Of these, **93.5%** are Case authors (*Case purity*), confirming that the pipeline predominantly isolates authors whose structural profiles deviate substantially from the Control norm. An author–subject enters the screen only when it has at least one identified non-self outgoing citation. The screen uses five measures of an author’s citation links: self-reference, citations to previous collaborators, mutual exchange, connections among neighbors, and concentration among recipients. These inputs capture author-edge cohesion or concentration. Same-journal referencing is excluded because it is defined on work references, citation balance because it requires internal incident weight, and year-to-year increase because it is temporal; all three are reported separately. All five screen inputs must be observed, and an unavailable value is not changed to zero.

Relative to non-outlier authors, Within each subject, every screen input is centered at

the Control median and divided by the Control interquartile range (IQR). A nonfinite or zero Control IQR is replaced by 1 only to avoid division by zero. Controls therefore define the preprocessing, detector fit, and thresholds; Cases do not affect those reference quantities.

490 No additional feature weights are applied, and the detected syndicate members exhibit many times higher cohesion signals (Fig. ??): **Clique Strength** $\approx 11\times$, **Co-author Citation Rate** $\approx 6.7\times$, **Reciprocity** $\approx 4.7\times$, and **Tunnel Vision (Outgoing HHI)** $\approx 3.1\times$. Note that the aggregate Case–Control fold-changes reported in Section ?? measure the same features across *all* matched pairs, whereas the fingerprint ratios here compare the 277 flagged outliers to the remaining population. These specific deviations constitute the unique *structural fingerprint* of the flagged cohesion outliers.

495 empirical percentile threshold makes no Gaussian assumption.

Outlier behavioural fingerprint. Radar chart of outlier fold-change ratios (log-scaled radial axis) relative to non-outlier authors across six citation-behaviour metrics. Values on each spoke give the outlier-to-normal mean ratio; concentric rings mark $1\times$, $10\times$, and $100\times$ baselines. The dashed inner ring corresponds to $1\times$ (no difference). **An Isolation Forest is a tree-based method that gives a higher score to a profile that is easy to separate**

500 **from typical Controls (F. T. Liu et al., 2008). A row passes this part of the screen when its score is above the 99th percentile of Controls in the same subject. A row also passes the second part when at least two of four cohesion measures exceed their Control percentiles. The final flag requires both parts. Because the four cohesion measures also enter the first part, the two parts are related and**

505 **are not independent evidence.**

Sensitivity analyses vary the Control percentile, random seed, and detector feature set. We report overlap between the detector and cohesion restriction, their rank association, and leave-one-feature-out stability. These quantities describe an operational screen; they are not measures of classification accuracy because

510 group is not a ground-truth anomaly label.

Table ?? demonstrates the superiority of the hybrid approach. While Isolation Forest alone achieves 84% purity, augmenting it with the cohesion filter improves purity to 94%. This indicates that approximately **Finally, we summarize**

where citation weight goes. The 2×2 table has the citing group in the rows and the cited group in the columns. Weighted assortativity compares the observed share of same-group citations with the share expected from the row and column totals (Newman, 2002):

$$r = \frac{\text{tr}(e) - \mathbf{a}^\top \mathbf{b}}{1 - \mathbf{a}^\top \mathbf{b}}. \quad (8)$$

Here e is the observed 2×2 table of citation-weight shares, a contains its row totals, b contains its column totals, and $\text{tr}(e)$ is the observed same-group share. Positive r indicates more same-group citation weight than expected from those totals. The reported label-swap p -value compares the observed same-group share with 10% of statistical outliers are legitimate prolific collaborators, who are successfully filtered out by the cohesion requirement, 1000 runs in which the two labels are swapped within each matched pair. The swaps keep the citation links and matching structure unchanged while removing the original group labels.

7.1 Network Forensics

525

8 Results

All numerical values in this section come from the same analysis and are shown in the tables and figures.

The largest outlier syndicate ($n = 23$) exhibits a distinct hub-and-spoke topology rather than a dense, uniform mesh (Figure ??). Within this group, the flow imbalance analysis reveals a clear hierarchy: a single “**Primary Beneficiary**” occupies the central position, receiving directed citation flows from peripheral “**Sycophants**” who define themselves by giving significantly more citations than they receive.

8.1 Matched differences in network measures

The primary and secondary matched comparisons are shown in Tables 3 and 4.

Table 3: Matched comparisons of the four primary measures.

Metric	Pairs	Case median	Control median	Median diff.	Boot. CI	Mean diff.	Mean boot. CI	Rank effect	Adj. <i>p</i>	Sign-flip <i>p</i>
Coauthor-citation rate	74,690	0.000	0.028	0.000	[0.000, 0.000]	0.048	[0.047, 0.050]	0.140	< 0.001	< 0.001
Reciprocity	39,441	0.000	0.000	0.000	[0.000, 0.000]	0.096	[0.092, 0.100]	0.341	< 0.001	< 0.001
Local clustering	102,626	0.000	0.000	0.000	[0.000, 0.000]	0.081	[0.079, 0.084]	0.264	< 0.001	< 0.001
Outgoing HHI	74,690	0.074	0.030	0.036	[0.035, 0.037]	0.097	[0.095, 0.099]	0.594	< 0.001	< 0.001

The Pairs column gives the number of matched pairs with both values present; this count can differ by metric. Each confidence interval (CI) is a 95% range from the paired bootstrap. The rank effect is the rank-biserial effect size. Median differences can be zero because many values are tied at zero, while mean differences show the average shift. Positive differences and positive rank-biserial effects indicate Case > Control. BH adjusts the four displayed *p*-values for multiple testing; the sign-flip *p*-value comes from randomly reversing pair directions.

Table 4: Matched comparisons of secondary measures.

Metric	Pairs	Case median	Control median	Median diff.	Boot. CI	Mean diff.	Mean boot. CI	Rank effect	Adj. <i>p</i>	Sign-flip <i>p</i>
Self-citation rate	75,526	0.000	0.007	0.000	[0.000, 0.000]	0.033	[0.031, 0.034]	0.177	< 0.001	< 0.001
Journal endogamy	79,700	0.008	0.064	-0.027	[-0.027, -0.026]	-0.009	[-0.010, -0.007]	-0.235	< 0.001	< 0.001
Within-sample citation balance	59,924	0.000	-0.024	0.000	[0.000, 0.000]	-0.020	[-0.028, -0.011]	-0.018	< 0.001	< 0.001
Maximum annual dyadic surge share	75,526	0.123	0.069	0.047	[0.046, 0.048]	0.094	[0.092, 0.096]	0.526	< 0.001	< 0.001

The Pairs column gives the number of matched pairs with both values present; this count can differ by metric. Each confidence interval (CI) is a 95% range from the paired bootstrap. The rank effect is the rank-biserial effect size. Median differences can be zero because many values are tied at zero, while mean differences show the average shift. Positive differences and positive rank-biserial effects indicate Case > Control. BH adjusts the four displayed *p*-values for multiple testing; the sign-flip *p*-value comes from randomly reversing pair directions.

535 Temporal activity supports the orchestration hypothesis, showing a sudden-burst pattern with high internal citation intensity.

As shown in the timeline analysis, citation accumulation is not organic but driven by coordinated waves, particularly evident in 2021 and 2024.

Largest outlier citation syndicate ($n = 23$). Force-directed layout of the internal directed citation network. Node size is proportional to betweenness centrality; = hub (highest betweenness); = net giver (out-degree > in-degree); = net receiver.

540 Arrows indicate the direction of citation flow; edge width scales with the number of citations between two authors. The inset box reports node count, directed edge count, and graph density.

The hub-and-spoke topology and coordinated temporal bursts reveal a pervasive “Sycophant-Beneficiary” flow imbalance that distinguishes Cases from Controls.

The four primary mean paired differences were Coauthor-citation rate 0.048,

95% bootstrap CI [0.047, 0.050]; Reciprocity 0.096, 95% bootstrap CI [0.092, 0.100]; Local clustering 0.081, 95% bootstrap CI [0.079, 0.084]; Outgoing HHI 0.097, 95% bootstrap CI [0.095, 0.099]. 3 median differences were zero. Outgoing HHI had a median paired difference of 0.036, 95% bootstrap CI [0.035, 0.037].

8.2 Case-Control Segregation

The median difference in same-journal referencing was -0.027, 95% CI [-0.027, -0.026]; the median difference in the maximum year-to-year increase to one recipient was 0.047, 95% CI [0.046, 0.048].

The distinction between low- and high-impact venues extends beyond simple density; it reflects extreme *segregation*. The high assortativity coefficient ($r = 0.71$) (Newman, 2002) and modularity ($Q = 0.97$) (Newman & Girvan, 2004) confirm a “**Two Worlds**” hypothesis, where Case authors operate in closed citation economies with minimal external engagement (Figure ??). Flagged outliers concentrate at the high-asymmetry, high-burst corner of the anomaly landscape, a region essentially unpopulated by Controls. In this sense, citation cliques are not merely organic clusters but isolated silos disconnected from the broader scientific discourse

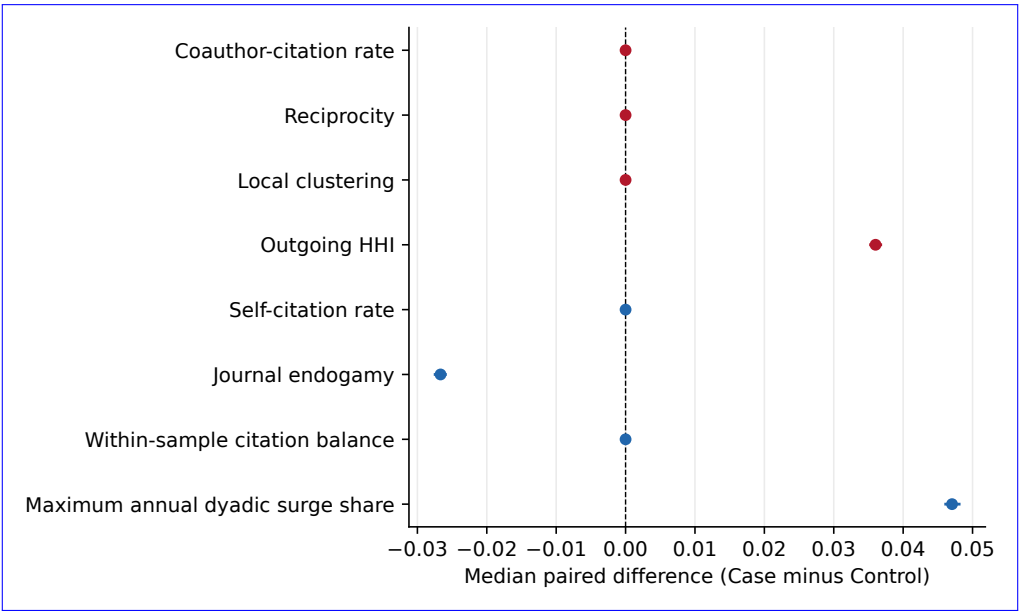


Figure 2: Matched effect estimates by outcome. Points show median paired differences and bars show paired-bootstrap 95% confidence intervals; mean differences and mean intervals are reported in Tables 3 and 4.

Table 5: Checks using exact h_5 matches and same-year collaborations.

Metric	Pairs	Case median	Control median	Median diff.	Boot. CI	Mean diff.	Mean boot. CI	Rank effect	Adj. p	Sign-flip p
Coauthor-citation rate	47,128	0.035	0.025	0.000	[0.000, 0.000]	0.075	[0.072, 0.077]	0.296	< 0.001	< 0.001
Reciprocity	24,032	0.140	0.000	0.015	[0.000, 0.031]	0.147	[0.142, 0.153]	0.470	< 0.001	< 0.001
Local clustering	64,176	0.000	0.000	0.000	[0.000, 0.000]	0.091	[0.088, 0.094]	0.330	< 0.001	< 0.001
Outgoing HHI	47,128	0.071	0.030	0.034	[0.033, 0.035]	0.097	[0.095, 0.099]	0.574	< 0.001	< 0.001
Coauthor-citation rate (same-year included)	74,690	0.068	0.080	0.000	[0.000, 0.000]	0.051	[0.049, 0.053]	0.105	< 0.001	< 0.001

The Pairs column gives the number of matched pairs with both values present; this count can differ by metric. Each confidence interval (CI) is a 95% range from the paired bootstrap. The rank effect is the rank-biserial effect size. The first four rows use exact- h_5 matches and BH correction across the primary outcomes. The same-year coauthor row uses the full cohort as a separate one-outcome check. A confidence interval (CI) gives the range supported by the paired bootstrap. Median differences can be zero because many values are tied at zero, while mean differences show the average shift.

Pair counts are reported separately because the missingness pattern differs

by measure. The count for every metric is checked not to exceed the 102,626 primary pairs.

Citation mixing matrix ($r = 0.71$, $Q = 0.97$; **strong homophily**). Row-normalised proportion of citations directed within and across tiers. Cell values show the conditional probability that an author in the row tier cites an author in the column tier. The Brewer RdBu colormap is centred at 0.5. Below: diagonal average (84.0%) summarises overall within-tier preference.

8.2 Sensitivity of primary comparisons

8.3 Cross-referencing Case Authors

To move from aggregate outlier statistics to actionable intelligence, we cross-reference the 277 hybrid outliers against real-world author identities and publication records. Of these, 189 (68.2%) had publicly resolvable ORCID profiles and are analysed below; the remaining 88 lacked public ORCID records or returned incomplete metadata. Each ORCID identifier is resolved to a name via the ORCID public API,¹ and publication portfolios are reconstructed from `impact.db` by joining `work_authors` to `works` and resolving journal titles through the `journal_names` catalogue. The exact- h_5 check had positive mean differences for all four primary comparisons; outgoing HHI had a median difference of 0.034 (95% CI [0.033, 0.035]). Using the same-year coauthor definition, the

¹

median difference was 0.000 (95% CI [0.000, 0.000]).

We define a composite *outlier score* as the weighted sum of population-normalised z -scores across six behavioural features: 575

$$S_i = \sum_{k=1}^6 w_k \cdot z_{ik}, \quad z_{ik} = \frac{x_{ik} - \bar{x}_k}{\sigma_k},$$

where x_{ik} is the raw value of feature k for author i , $\bar{x}_k$ and σ_k are the population mean and standard deviation, and the weights w_k rank the six features by their discriminative strength between Case and Control authors. To avoid overfitting the weights to the sample, we did not optimise them numerically; instead, each feature was assigned an integer or half-integer weight on the grid $\{2.0, 3.0, 3.5, 4.0\}$ in proportion to two complementary evidence sources: (i) Random-Forest mean decrease in impurity for Case–Control classification (Figure ??), and (ii) the Wilcoxon rank-biserial correlation from the matched-pair tests. Features 580 ranking top on *both* criteria received the largest weight; features ranking high on one criterion only, or significant with a smaller effect size, received a lower weight. The resulting schedule is: co-author citation rate ($w = 4.0$; largest effect size and top RF importance), clique strength (3.5) and reciprocity rate (3.5; large effects, second-tier RF importance), outgoing HHI (3.0; concentration signal complementary to the cohesion metrics), self-citation rate (2.0) and journal endogamy rate (2.0; smaller but still significant Case–Control differences). The score is a heuristic ranking device rather than a fitted model; the final outlier flag 585 is issued only when S_i and the independent Isolation Forest both exceed their thresholds.

8.3 Control-referenced anomaly enrichment

Authors are ranked by S_i in descending order. An individual feature is flagged when its z -score exceeds 3σ or 5σ . The screen flagged 1880 Cases and 162 Controls (2.48% versus 0.16%; 15.26-fold difference; Fisher exact $p < 0.001$). 590

Key Findings. Of the 189 outliers, 162 (85.7%) belong to the Case group, confirming the precision of the Isolation Forest detector. Outliers concentrate in Health Sciences ($n = 77$) and Physical Sciences ($n = 71$), followed by Social Sciences ($n = 46$) and Life Sciences ($n = 3$); per-subject counts sum to 197 because eight authors are flagged in two subjects each. A total of 131 outliers (69.3%) belong to at least one syndicate (connected component among mutual outlier citation links), with the largest syndicate comprising 23 members (see Figure ??). 595

Table ?? presents the ten highest-scoring authors. All ten exhibit co-author citation rates exceeding 5σ ; seven also show

Table 6: Screen results by journal-impact tier.

Tier	Eligible	Screenable	Flagged	Flagged share	Case/Control enrichment	Case share of flags	Fisher exact p
Case	75,954	53,365	1,880	2.48%	15.26	92.07%	< 0.001
Control	99,889	71,146	162	0.16%	15.26	–	< 0.001

Eligibility requires identified non-self outgoing citation weight. Enrichment is the Case flagged share divided by the Control flagged share. The Fisher exact p -value tests whether the two flagged shares differ. Flags are screening results and do not establish intent or misconduct.

Table 7: The two screen rules and their overlap.

Detector exceeds threshold	Cohesion rule: no	Cohesion rule: yes	Total
No	118,525 (95.19%)	14 (0.01%)	118,539 (95.20%)
Yes	3,930 (3.16%)	2,042 (1.64%)	5,972 (4.80%)
Total	122,455 (98.35%)	2,056 (1.65%)	124,511 (100.00%)

Cells are counts (percent of 124,511 screenable author–subject rows). Spearman’s ρ between detector score and cohesion exceedance count was 0.421 ($n = 124,511$; $p < 0.001$). The four cohesion measures are detector inputs, so this is an overlapping interpretable restriction rather than independent corroboration.

reciprocity rates beyond 5σ . The two top-ranked authors — both in agricultural engineering — share an identical outlier score ($S = 658.6$) and belong to the same 23-member syndicate, consistent with coordinated behaviour. The two screen rules agreed for 2,042 rows; 3,930 passed only the first rule and 14 only the second. Their scores were positively related ($\rho = 0.421$, $p < 0.001$).

Removing one input at a time changed little: flag overlap ranged from 0.931 to 0.995, and Case/Control enrichment ranged from 14.83 to 15.58. Details appear in Table 12.

Table 8: Sensitivity of screen results.

Control percentile	Flagged count: median [range]	Case share	Control share	Enrichment	Overlap with main screen
97.5%	3968 [3942–3985]	4.67%	0.42%	11.09	0.51
99.0%	2046 [2029–2052]	2.48%	0.16%	15.25	1.00
99.5%	1213 [1180–1224]	1.50%	0.07%	20.46	0.59

Each row summarises ten fixed seeds; the main screen uses the 99th percentile and seed 42. Enrichment is the Case flagged share divided by the Control flagged share. Overlap is the Jaccard share of flagged rows retained by both screens.

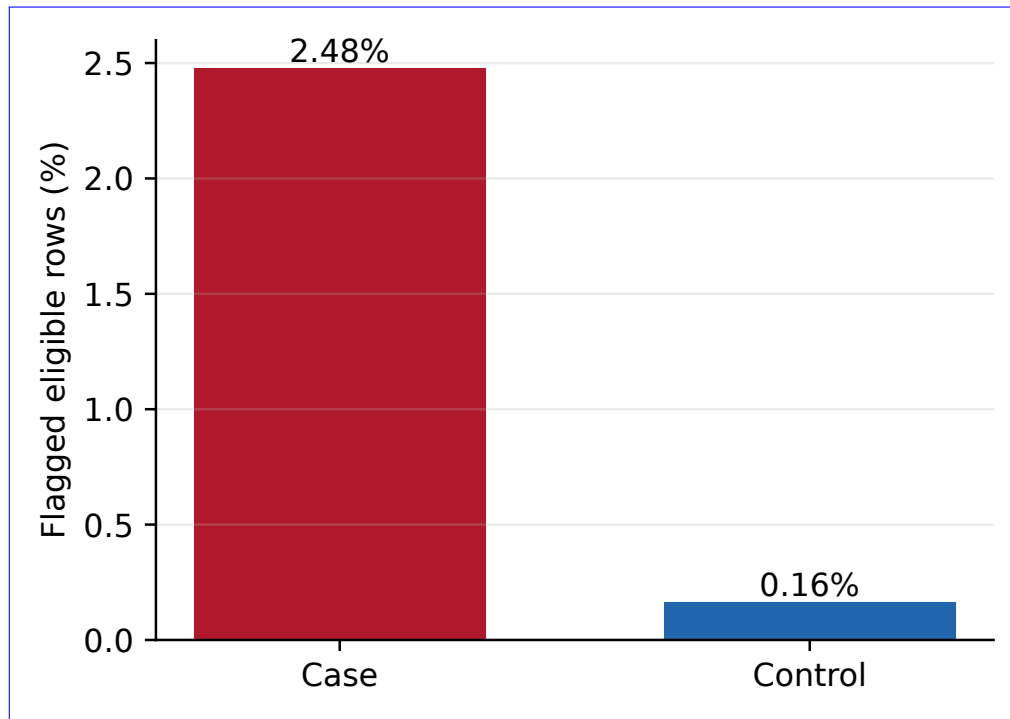


Figure 3: Share of eligible author-subjects flagged in each group under the declared screen. Values and supporting counts are generated from the same final flag used in all downstream analyses.

8.4 Fractional group mixing

94.03% of matched citation weight stayed within the same group (label-swap 605 $p < 0.001$).

Table 9: Where matched authors cite.

Citing group	Cited Case	Cited Control
Case	27500.67	1765.51
Control	431.67	7104.63

Cells are cumulative fractional citation weights. The within-tier share is 94.03%, and weighted assortativity is 0.828; this compares same-tier weight with the amount expected from the row and column totals. The two-sided 10,000 within-pair tier-label-swap p -value is < 0.001 .

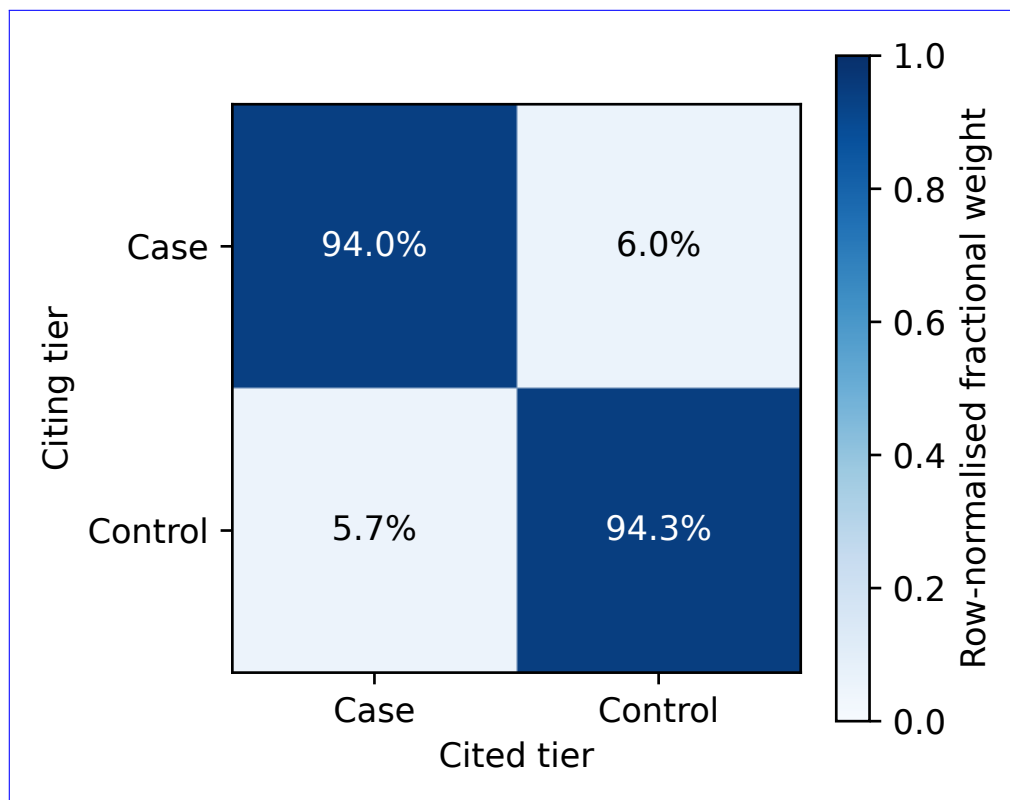


Figure 4: Fractional citation-weight mixing between Case and Control nodes. The comparison distribution is based on within-pair group-label swaps.

8.5 Formal reciprocal-clique structure

Primary-rule reciprocal-clique membership was 10.96% for Cases and 3.82% for Controls (paired difference 7.14 percentage points; exact $p < 0.001$).

Table 10: Matched reciprocal-clique membership comparison.

Primary rule	All groups	Recip. groups	Case members	Control members	Paired diff.	Case only	Control only	Exact p
$k \geq 4, r \geq 0.50$	36,311	9,362	11,252 3,925 (3.82%) (10.96%)		7.14 pp	10,678	3,351	< 0.001

A candidate group is a maximal group in which every pair has a citation in at least one direction. The primary rule requires at least four members, density at least 0.75, and weighted reciprocity at least 0.50. Authors in overlapping qualifying groups are counted once per subject. The exact two-sided matched test uses only discordant pairs.

Table 11: Sensitivity of the clique rule.

Minimum group size	Recip. threshold	Recip. groups	Case rate	Control rate	Paired diff.	Case only	Control only	Exact p	Subjects + / - / =
3	0.25	33,041	21.11%	9.89%	11.22 pp	19,007	7,491	< 0.001	5/0/0
3	0.50	10,237	12.21%	4.82%	7.39 pp	11,770	4,190	< 0.001	5/0/0
3	0.75	1,420	2.85%	1.25%	1.60 pp	2,879	1,238	< 0.001	5/0/0
4	0.25	31,621	19.47%	8.21%	11.26 pp	17,865	6,311	< 0.001	5/0/0
4	0.50	9,362	10.96%	3.82%	7.14 pp	10,678	3,351	< 0.001	5/0/0
4	0.75	982	2.05%	0.82%	1.22 pp	2,080	823	< 0.001	4/1/0
5	0.25	25,836	14.62%	3.99%	10.63 pp	14,145	3,238	< 0.001	4/0/1
5	0.50	6,677	8.03%	1.82%	6.21 pp	8,020	1,648	< 0.001	4/0/1
5	0.75	443	1.18%	0.35%	0.83 pp	1,213	361	< 0.001	3/1/1

The directed density threshold is fixed at 0.75. The reciprocity threshold is the minimum share of two-way citation weight. Membership is unique by author–subject, and the last column reports subjects with higher Case rates, higher Control rates, or ties.

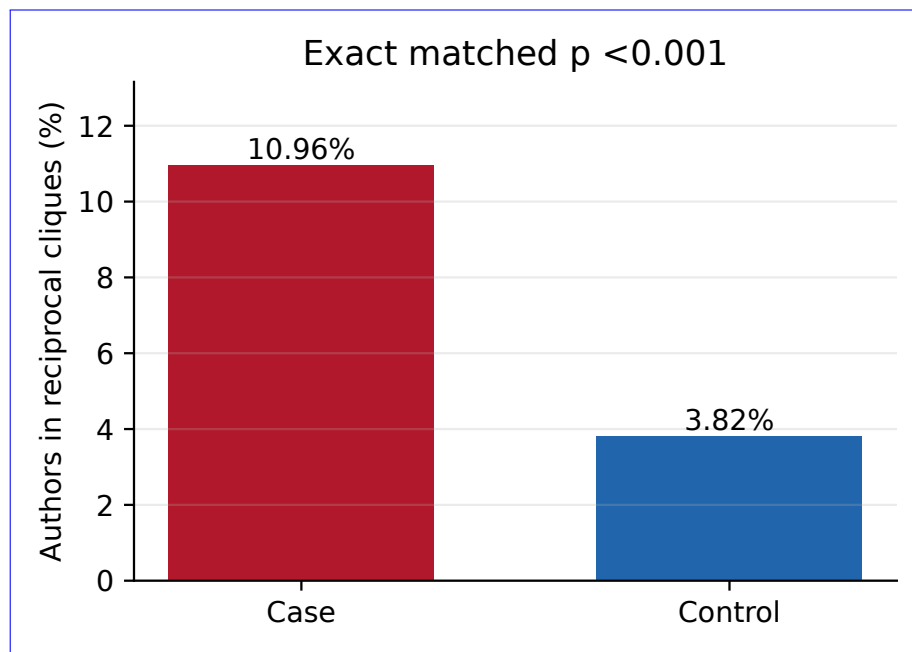


Figure 5: Unique author–subject membership rates in primary-rule reciprocal citation cliques. Authors in overlapping cliques are counted once.

Here, a clique is a group of at least four matched authors in which every 610

pair is connected by a citation in at least one direction. The presence of authors writing in Korean, Russian, Ukrainian, and Portuguese underscores the international scope of the phenomenon. primary inference compares the binary author–subject membership indicator within pairs rather than counting repeated appearances in overlapping cliques. There were 10678
615 pairs in which only the Case belonged and 3351 in which only the Control belonged; the exact matched result was $p < 0.001$. The subject-level breakdown is reported with every threshold sensitivity.

Table ?? provides a publication-level audit of the same ten authors . Typical portfolios are small (4–14 works, median 5), concentrated in a single regional journal, and characterised by very few distinct citation partners (1–5) yet elevated reciprocal
620 citation links, a pattern consistent with closed-loop mutual citation within a small community of co-authors.

9 Discussion

This study compares two journal-impact groups after matching authors within subject and on recent citation productivity. The result is an association within matched pairs. It is not an estimate of what would happen if an author changed
625 venues, because venue choice, collaboration, topic, and citation practice can affect one another.

The four primary mean paired differences were Coauthor-citation rate 0.048, 95% bootstrap CI [0.047, 0.050]; Reciprocity 0.096, 95% bootstrap CI [0.092, 0.100]; Local clustering 0.081, 95% bootstrap CI [0.079, 0.084]; Outgoing HHI
630 0.097, 95% bootstrap CI [0.095, 0.099]. 3 median differences were zero. Outgoing HHI had a median paired difference of 0.036, 95% bootstrap CI [0.035, 0.037]. This comparison complements the author-level concentration measures of Evdaimon et al., while measuring the distribution of references sent by an author rather than the number of authors needed to account for received citations (Evdaimon,
635 Ioannidis, et al., 2024).

The concentration of outlier behaviour in small, regional journals many of which do not appear in major indexing databases, suggests that coordinated citation inflation is disproportionately prevalent in venues with limited editorial oversight. The publication-audit data show that the highest-scoring authors maintain very small portfolios (median 5 works), yet achieve inflated citation metrics through closed-loop mutual referencing. This pattern is consistent with prior work on “citation stacking” (Heneberg, 2016) and suggests that secondary results add detail. The median difference in same-journal referencing was -0.027, 95% CI [-0.027, -0.026]; the median difference in the maximum year-to-year increase to one recipient was 0.047, 95% CI [0.046, 0.048]. These measures describe different parts of citation behavior from the journal-level surveillance of incoming self-citation rates from small author cliques could serve as an effective early-warning signal citation stacking studied by Heneberg (Heneberg, 2016). The screen flagged 2.48% of Case rows and 0.16% of Control rows, a 15.26-fold difference. 94.03% of matched citation weight stayed within the same group (label-swap $p < 0.001$). These are descriptive results that show how the groups differ; they do not identify a cause.

While broad asymmetry metrics are generally subdued, the internal structure of detected syndicates reveals a distinct flow imbalance. As seen in the largest syndicate ($n = 23$), a hub-and-spoke topology emerges where peripheral members (“Sycophants”) act as net givers to central “Beneficiaries”. This suggests that the observed cohesion serves a functional purpose: the coordinated funneling of credit to central figures, rather than reciprocal exchange among equals. The clique analysis makes the main pattern more specific. Primary-rule reciprocal-clique membership was 10.96% for Cases and 3.82% for Controls (paired difference 7.14 percentage points; exact $p < 0.001$). Membership is defined once per author–subject even when cliques overlap, and the exact test preserves the matched design. The evidence therefore concerns which matched authors belong to the defined structure, while leaving the reasons for the pattern open.

These findings extend the concept of *citation orchestration* (Evdaimon, Ioannidis, et al., 2024), showing that orchestration manifests as persistent reciprocity and dense closure. The limited cross-tier mixing supports concerns about *home bias* (Qiu et al., 2025) and institutional siloing. However, the specificity of our *hybrid detection* (93.5% purity) suggests that these patterns

in low-impact venues are structural anomalies distinct from legitimate topic specialization. Kojaku et al. detect anomalous groups of journals by comparing journal-level citation structure with network expectations (Kojaku et al., 2021). Our result instead compares author-level clique membership within matched pairs under a declared structural rule. The approaches share a relational perspective, but their populations, estimands, and null comparisons differ, so their estimates are not numerically interchangeable and neither structural result alone establishes misconduct.

The results highlight that aggregate impact metrics hide structural manipulation. Evaluation systems must incorporate **cohesion-aware metrics** — specifically *co-author citation rates* and *flow imbalance*— to distinguish organic collaboration from dense internal referencing. The success of the hybrid pipeline suggests that combining multivariate anomaly detection (Isolation Forest) with targeted cohesion signals can effectively flag insular publication ecosystems. The screen and clique rule are measurement tools for follow-up. They summarize citation links and cannot determine whether a citation was appropriate, coordinated, or intentional. Narrow specialties, stable research teams, and institutional or career-stage differences can also create cohesive citation groups.

Overall, the dominance of segregation and flow imbalance suggests that low-impact venues foster **closed economies** of citation. Understanding these patterns is essential for designing fair bibliometric indicators that reward genuine impact rather than inward-looking visibility loops.

10 Limitations and Future Work

10 Limitations

This study relies on Crossref metadata (The estimates apply to the fixed 2020–2024), which vary in completeness. While the hybrid filter significantly reduces false positives, the distinction between a “cartel” and a hyper-specialized research group remains a probabilistic assessment. The LLM-based journal classification (Appendix ??) was not validated against an independent ground truth; misclassification of some journals could shift the boundary between Case and

Control authors . Additionally, journal classification relied solely on a proprietary model (gpt-4.1-2025-04-14) without an open-weight baseline comparison and without independent human validation of the resulting labels (LLM Guidelines Consortium, 2024). Moreover, citation links are analyzed structurally, without text-level context to infer the motivation of citing (J. Liu et al., 2024). **snapshot and the author–subject observations that meet the stated inclusion criteria. The graph metrics therefore describe the defined analytic population and graph construction.**

Ethical considerations. Where individual author profiles are discussed (Tables ?? and ??), the underlying data consist exclusively of publicly available Crossref metadata and self-declared ORCID identifiers. Identifying anomalous citation patterns in publicly funded research serves a legitimate accountability function, consistent with established practice in the retraction and research-integrity literature (Besançon, Cabanac, et al., 2024; Biagioli & Lippman, 2020). Although explicit author names and institutional affiliations are omitted from the text and tables for discretion, the underlying publication records remain fully transparent and verifiable through the provided ORCID identifiers.

Future work could combine the structural fingerprint with author disambiguation and profile-level inspection to move from structural suspicion to targeted auditing. Extending the temporal coverage could clarify whether the observed “Two Worlds” segregation is widening over time. Experimental audits in selected venues could also test whether editorial interventions can dismantle the echo chambers identified in this analysis.

11 Summary and Conclusions

We developed a subject-aware hybrid pipeline that integrates LLM-based journal classification, per-subject Eigenfactor scores, and a composite outlier detection strategy. This design allowed us to isolate behavioral signatures with author productivity (by The five broad subject categories cannot capture fine topical proximity. The journal classifier was not independently validated against blinded human coding, and the publication-weighted Eigenfactor cutoffs remain sensitive to the observed journal network. ORCID availability, the 70% portfolio rule, and the positive- h_5 matching pool define the study population. Matching on subject-specific h_5) held constant improves comparability within that pool but does not balance geography,

language, institution, team structure, career age, or unmeasured research topic.

The evidence points to a highly structured phenomenon. In the aggregate Case–Control comparison, authors publishing predominantly in low-impact journals cite co-authors **6.7× more frequently**; among the 277 flagged outliers the lift is even starker, with **11× greater clique strength** relative to non-outliers. Rather than random noise, outlier syndicates display a **hub-and-spoke topology**, where peripheral “Sycophants” funnel citations to central “Beneficiaries” through coordinated temporal citation bursts. The flow imbalance analysis confirms that Case authors systematically give more citations than they receive. Cumulative citation trajectories over the **Graph metrics depend on population boundaries**. Ego-layer measures include identified recipients outside the matched cohort, whereas reciprocity, clustering, balance, and mixing are restricted to matched nodes. Results from the latter should not be interpreted as full-network properties. The 2020–2024 window suggest that the gap between tiers may be widening, although a longer observation period is needed to confirm this trend. citation window limits observed citation behavior, while historical work and author metadata in the snapshot can left-censor or misdate prior collaborations. The year dimension of the edge data supports coauthor timing and the scalar surge-share outcome, but is not analyzed as a group-specific trajectory or group-by-year interaction; we therefore make no claim that a Case–Control gap widens over time.

Controlling for subject and productivity, the hybrid detection strategy identifies structurally anomalous citation groups with 93.5% Case purity (proportion of flagged authors belonging to the Case group). These groups are characterized not by simple asymmetry, but by a specific cohesion fingerprint: extreme co-author citation rates, dense local clustering, and tunnel vision. Consequently, some low-impact venues appear to foster **segregated, inward-looking citation economies** whose structural properties are associated with distortion of bibliometric indicators. The anomaly thresholds and Isolation Forest define an operational screen, not a validated diagnostic. Percentile, seed, and leave-one-feature-out sensitivity quantify some model dependence but do not supply ground-truth labels or make the overlapping restriction independent. Finally, structural metadata cannot reveal the textual function, scientific relevance, or motivation of a citation (J. Liu et al., 2024).

11 Conclusion

All four primary mean differences favored Cases; all four comparisons were significant after adjustment, and 3 median differences were zero.

Lower-impact-group authors were more likely than matched higher-impact-group authors to belong to formally defined reciprocal citation cliques. The paired membership difference was 7.14 percentage points (exact $p < 0.001$). This is an associational result for the defined cohort and supplies a pattern for follow-up without assigning a cause or intent.

12 Author Contributions

Conceptualization: P.-A.S., D.S.; Methodology: P.-A.S.; Software: P.-A.S., G.A.; Data curation: P.-A.S., G.A.; Validation: P.-A.S., G.A., D.S.; Formal analysis: P.-A.S., G.A.; Visualization: P.-A.S.; Writing—original draft: G.A.; Writing—review & , review and editing: P.-A.S., G.A., D.S.; Supervision: D.S.; Project administration: D.S.

13 Competing Interests

The authors declare no conflict of interest competing interests.

14 Funding Information

This research did not receive a specific grant from any funding agency in the public, commercial or non-profit , or not-for-profit sectors.

15 Data and Code Availability

All essential data and code for this study are openly available at Zenodo under a CC BY license

The v4.0.0 software and aggregate replication materials for this revision are archived on Zenodo (Spanakis et al., 2026). The analysis is implemented in a single, self-contained script (`citation_analysis.py`) that reproduces all results in one invocation: it loads the pre-built `rolap.db` and `impact.db` databases, runs the statistical comparisons (Section ??), generates all eight publication figures, and produces the author outlier tables (Section ??). The citation databases and the derivative tables can be reproduced through the 2024 Crossref public data set by means of the supplied Unix Makefile and SQL scripts. **fixed raw snapshot is not redistributed.**

References

- Abalkina, A., Aquarius, R., Bik, E. M., Bimler, D., Bishop, D., Byrne, J., Cabanac, G., Day, A., Labbé, C., & Wise, N. (2025). Stamp out paper mills. *Nature*, *637*, 1047–1049. <https://doi.org/10.1038/d41586-025-00212-1>
- Aksnes, D. W. (2003). A macro study of self-citation. *Scientometrics*, *56*(2), 235–246. <https://doi.org/10.1023/A:1021919228368>
- Baccini, A., De Nicolao, G., & Petrovich, E. (2019). Citation gaming induced by bibliometric evaluation: A country-level comparative analysis. *PLOS ONE*, *14*(9), e0221212. <https://doi.org/10.1371/journal.pone.0221212>
- Beall, J. (2012). Predatory publishers are corrupting open access. *Nature*, *489*, 179. <https://doi.org/10.1038/489179a>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, *57*(1), 289–300.
- Bergstrom, C. T., West, J. D., & Wiseman, M. A. (2008). The eigenfactor metrics: A network approach to assessing scholarly journals. *Journal of Neuroscience*, *28*(45), 11433–11434. <https://doi.org/10.1523/JNEUROSCI.0003-08.2008>

- Besançon, L., Cabanac, G., et al. (2024). When scientific citations go rogue: Uncovering sneaked references. *Journal of the Association for Information Science and Technology*. <https://doi.org/10.1002/asi.24896>
- Biagioli, M., Kenney, M., Martin, B. R., & Walsh, J. P. (2019). Academic misconduct, 785 misrepresentation and gaming: A reassessment. *Research Policy*, 48(2), 401–413. <https://doi.org/10.1016/j.respol.2018.10.025>
- Biagioli, M., & Lippman, A. (Eds.). (2020). *Gaming the metrics: Misconduct and manipulation in academic research*. MIT Press. <https://doi.org/10.7551/mitpress/11087.001.0001>
- Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding 790 of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, (10), P10008.
- Breunig, M. M., Kriegel, H.-P., Ng, R. T., & Sander, J. (2000). LOF: Identifying density-based local outliers. *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data*, 93–104. <https://doi.org/10.1145/342009.335388> 795
- Burnham, J. F. (2006). Scopus database: A review. *Biomedical Digital Libraries*, 3(1), 1. <https://doi.org/10.1186/1742-5581-3-1>
- Catanzaro, M. (2024). Citation manipulation found to be rife in math. *Science*, 383(6682), 470. <https://doi.org/10.1126/science.ado3859>
- Christopher, J. (2021). The raw truth about paper mills. *FEBS Letters*, 595(13), 1751–1757.800 <https://doi.org/10.1002/1873-3468.14143>
- Cliff, N. (1993). *Dominance statistics: Ordinal analyses to answer ordinal questions*. Psychological Assessment Resources.
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd Edition). Routledge. 805
- Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman & Hall.

- Evdaïmon, I., Ioannidis, J. P. A., et al. (2024). Metrics to detect small-scale and large-scale citation orchestration. *arXiv Cornell University*. <https://doi.org/10.48550/arXiv.2406.19219>
- 810 Fister Jr., I., Fister, I., & Perc, M. (2016). Toward the discovery of citation cartels in citation networks. *Frontiers in Physics*, 4, 49. <https://doi.org/10.3389/fphy.2016.00049>
- Garfield, E. (1972). Citation analysis as a tool in journal evaluation. *Science*, 178(4060), 471–479. <https://doi.org/10.1126/science.178.4060.471>
- Garfield, E. (2006). The history and meaning of the journal impact factor. *Journal of the*
815 *American Medical Association*, 295(1), 90–93.
- González-Pereira, B., Guerrero-Bote, V. P., & Moya-Anegón, F. (2010). A new approach to the metric of journals’ scientific prestige: The SJR indicator. *Journal of Informetrics*, 4(3), 379–391. <https://doi.org/10.1016/j.joi.2010.03.002>
- Goodhart, C. A. E. (1984). Problems of monetary management: The UK experience [Originally
820 published in 1975 by the Reserve Bank of Australia under “Papers in monetary economics”]. In *Monetary theory and practice* (pp. 91–121). Macmillan Education UK. https://doi.org/10.1007/978-1-349-17295-5_4
- Heneberg, P. (2016). From excessive journal self-cites to citation stacking: Analysis of journal self-citation kinetics in search for journals, which boost their scientometric indicators.
825 *PLoS ONE*, 11(4), e0153730. <https://doi.org/10.1371/journal.pone.0153730>
- Hicks, D., Wouters, P., Waltman, L., de Rijcke, S., & Rafols, I. (2015). Bibliometrics: The Leiden Manifesto for research metrics. *Nature*, 520(7548), 429–431. <https://doi.org/10.1038/520429a>
- Hirsch, J. E. (2005). An index to quantify an individual’s scientific research output. *Proceedings*
830 *of the National Academy of Sciences*, 102(46), 16569–16572. <https://doi.org/10.1073/pnas.0507655102>
- Hirschman, A. O. (1964). The paternity of an index. *The American Economic Review*, 54(5), 761–762.

- Ioannidis, J. P. A. (2025). Features and signals in precocious citation impact: A meta-research study. *PLOS ONE*, 20(8), e0328531. <https://doi.org/10.1371/journal.pone.0328531> 835
- Ioannidis, J. P. A., Baas, J., Klavans, R., & Boyack, K. W. (2019). A standardized citation metrics author database annotated for scientific field. *PLOS Biology*, 17(8), e3000384. <https://doi.org/10.1371/journal.pbio.3000384>
- Ioannidis, J. P. A., Collins, T. A., & Baas, J. (2024). Evolving patterns of extreme publishing behavior across science. *Scientometrics*, 129, 5783–5796. <https://doi.org/10.1007/s11192-024-05117-w> 840
- Kojaku, S., Livan, G., & Masuda, N. (2021). Detecting anomalous citation groups in journal networks. *Scientific Reports*, 11(1), 14524. <https://doi.org/10.1038/s41598-021-93572-3>
- Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation forest. *Proceedings of the 8th IEEE International Conference on Data Mining*, 413–422. 845
- Liu, J., Bai, X., Wang, M., Tuarob, S., & Xia, F. (2024). Anomalous citations detection in academic networks. *Artificial Intelligence Review*, 57(4), 103. <https://doi.org/10.1007/s10462-023-10655-5>
- LLM Guidelines Consortium. (2024). *LLM Guidelines: Transparency in LLM-assisted research*. 850 Retrieved September 17, 2025, from <https://llm-guidelines.org/>
- Naddaf, M. (2025). How to spot fake scientists and stop them from publishing papers. *Nature*, 646, 792–794. <https://doi.org/10.1038/d41586-025-03341-9>
- Newman, M. E. J. (2002). Assortative mixing in networks. *Physical Review Letters*, 89(20), 208701. <https://doi.org/10.1103/PhysRevLett.89.208701> 855
- Newman, M. E. J., & Girvan, M. (2004). Finding and evaluating community structure in networks. *Physical Review E*, 69(2), 026113. <https://doi.org/10.1103/PhysRevE.69.026113>

- Perianes-Rodriguez, A., Waltman, L., & van Eck, N. J. (2016). Constructing bibliometric
 860 networks: A comparison between full and fractional counting. *Journal of Informetrics*,
 10(4), 1178–1195. <https://doi.org/10.1016/j.joi.2016.10.006>
- Qiu, S., Steinwender, C., & Azoulay, P. (2025). *Paper tiger? Chinese science and home bias
 in citations* (NBER Working Paper No. 32468) (Revised from May 2024 version).
 National Bureau of Economic Research. <https://doi.org/10.3386/w32468>
- 865 Spanakis, P.-A., Alexandrou, G., & Spinellis, D. (2026). *Replication package for: Citation-
 network cohesion and reciprocal citation cliques across journal-impact groups: A
 matched author analysis* (Version v4.0.0). Zenodo. <https://doi.org/10.5281/zenodo.19786936>
- Spinellis, D. (2023). Open reproducible scientometric research with Alexandria3k. *PLoS ONE*,
 870 18(11), e0294946. <https://doi.org/10.1371/journal.pone.0294946>
- Spinellis, D. (2024a). Unit tests for SQL. *IEEE Software*, 41(1), 31–34. <https://doi.org/10.1109/MS.2023.3328788>
- Spinellis, D. (2024b). Engineering data processing workflows. *IEEE Software*, 41(4), 25–29.
<https://doi.org/10.1109/MS.2024.3385665>
- 875 Spinellis, D. (2025). False authorship: An explorative case study around an AI-generated
 article published under my name. *Research Integrity and Peer Review*, 10(1). <https://doi.org/10.1186/s41073-025-00165-z>
- Van Noorden, R., & Chawla, D. S. (2019). Hundreds of extreme self-citing scientists revealed
 in new database. *Nature*, 572(7771), 578–579. [https://doi.org/10.1038/d41586-019-02](https://doi.org/10.1038/d41586-019-02479-7)
 880 [479-7](https://doi.org/10.1038/d41586-019-02479-7)
- Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6),
 80–83.
- Wilhite, A. W., & Fong, E. A. (2012). Coercive citation in academic publishing. *Science*,
 335(6068), 542–543. <https://doi.org/10.1126/science.1212540>

A Feature Definitions Upstream Venue-Score Specification

885

Table ?? provides formal definitions for all behavioural metrics used in the analysis. In the formulas below, w_{ij} denotes the fractional citation weight from author i to author j . For subject s , let $Z_{jk}^{(s)}$ count citations from citing journal j (see Appendix ??), and sums run over non-self edges unless stated otherwise.

B Matching Procedure and Detection Parameters

A.1 Author Classification

890

Authors were classified based on their publication portfolio: **Case:** $\geq 70\%$ of papers published in journals below the 25th to cited journal k . The diagonal is removed, and each non-dangling citing-journal column of H_s is

$$(H_s)_{kj} = \frac{Z_{jk}^{(s)}}{\sum_{\ell} Z_{j\ell}^{(s)}}. \quad (9)$$

Let a_s be the normalized journal article-count vector and d_s indicate dangling columns. With damping $\alpha = 0.85$, the subject-specific Eigenfactor percentile, with a minimum of 3 papers. **Control:** $\geq 70\%$ of papers published in journals above the 75th subject-specific Eigenfactor percentile, with a minimum of 3 papers. **Other:** authors not meeting either threshold. influence vector is obtained by

$$\pi_s = \alpha H_s \pi_s + [\alpha d_s^T \pi_s + (1 - \alpha)] a_s. \quad (10)$$

The received-influence vector $H_s \pi_s$ is normalized to sum to 100 within subject. Thus citation mass flows from citing to cited journals, journal self-citations are excluded, and dangling mass and restart mass follow article counts. These fixed journal-level scores are upstream inputs to the author matching and are not author-network outcomes.

B Matching and Sensitivity Specifications

Cases and Controls require a non-empty ORCID, at least three eligible works,

905 and a 70% threshold ensures that classified authors are predominantly active in one venue stratum while tolerating occasional cross-tier publications (e.g. a single paper in a high-impact journal by an otherwise low-tier author). Lower thresholds (50–60%) admit too many mixed-portfolio authors, diluting the behavioural contrast; higher thresholds ($\geq 80\%$) reduce sample sizes substantially and bias the cohort toward single-journal authors. The 70% cut-off balances purity against statistical power and is consistent with the majority-venue classification used in large-scale author-level analyses (Ioannidis et al., 2019, 2024).

910 B.1 Matched-Pair Construction

Case–Control pairs were formed via greedy one-pass matching within each subject area. For each Case author the algorithm selected the closest available Control author by subject-specific portfolio share in the relevant journal group. The matching pool additionally requires a positive h_5 index. Productivity was bucketed by integer division ($b = \lfloor h_5/3 \rfloor$); candidates with $|b_{\text{case}} - b_{\text{control}}| \leq 1$ (i.e., ± 3 value; zero- h_5 points) were 915 eligible. Among eligible candidates the one minimising $|h_{5,\text{case}} - h_{5,\text{control}}|$ was selected. Each (ORCID, subject) pair was used at most once (matching without replacement). This procedure yielded 9 431 matched pairs across five subject areas (Table ??). profiles are not candidates. Within that pool, candidate eligibility uses the inclusive hard caliper $|h_{5,\text{Case}} - h_{5,\text{Control}}| \leq 3$. Greedy selection proceeds without replacement within subject in the total order subject, ascending absolute h_5 920 distance, Case ORCID, and Control ORCID. The generated cohort must pass the same subject-keyed positive- h_5 join and caliper checks. Exact- h_5 pairs form a sensitivity subset rather than a replacement cohort.

B.1 Fractional Citation Weights

Each citation edge between a citing work with n_c authors and a cited work with n_a authors received weight

$$w = \frac{1}{n_c \times n_a}.$$

This fractional counting scheme (Perianes-Rodriguez et al., 2016) down-weights edges from heavily co-authored papers, preventing a single multi-author article from dominating an individual’s citation profile. Self-loops ($i = j$) were retained and tracked via a Boolean flag for the self-citation rate metric ; all other metrics were computed on non-self edges. The temporal window covers works published between 2020 and 2024 inclusive, yielding 2 525 512 weighted citation edges (2 507 806 non-self, 17 706 self-citations) among 18 862 study authors.

B.1 Outlier Detection

930

Table ?? summarises the Isolation Forest hyper-parameters. The model was trained per-subject on 13 of the 14 scaled features listed in Table ?? (excluding normalised burst intensity; see Section ??) **The paired bootstrap resamples pair identifiers with replacement. The sign-flip test acts on complete-pair differences for one metric at a time. Multiple-testing adjustment is not pooled across primary and secondary families. The same-year coauthor sensitivity changes only $c_{ij} < t$ to $c_{ij} \leq t$; all other definitions and pair membership remain fixed.**

C Anomaly-Screen Specification

Cohesion composite score. A secondary filter computes a weighted z -score relative to the Control population:

$$S_i = \sum_{k=1}^6 w_k \cdot z_{ik}, \quad z_{ik} = \frac{x_{ik} - \bar{x}_k}{\sigma_k},$$

with weights: co-author citation rate (4.0), clique strength (3.5), reciprocity rate (3.5), outgoing HHI (3.0), self-citation rate (2.0), journal endogamy (2.0). Standardisation uses the population mean and standard deviation. An author was flagged only if both the Isolation Forest anomaly score exceeded its per-subject threshold *and* $S_i > 4\sigma$.

D Data Derivation and Sampling

The analytical dataset was derived through a multi-stage pipeline. This appendix documents the full data flow and the filtering decisions at each stage.

945 C.1 Stage 1: Raw Data Ingestion

Crossref metadata were ingested via *alexandria3k* (Spinellis, 2023), retaining works published between 2020 Controls define subject-specific robust centers, scales, detector fits, score cutoffs, and 2024. The following fields were extracted: DOI, publication year, page information, print and electronic ISSN, reference lists (DOI-level), and author ORCID identifiers. univariate cohesion cutoffs. Cases do not affect 950 those reference quantities. The canonical feature output retains the detector score, cohesion-restriction indicator, citation eligibility, and final flag for each (ORCID, s , tier) row. The downstream mixing routine consumes that final flag rather than recomputing or substituting a label.

C.1 Stage 2: Journal-Level Metrics

955 Works were linked to journals through normalised ISSNs. A citability filter removed non-article items (single-page entries such as editorials and news pieces). For each journal, two-year The primary screen uses seed 42 and the 99th Control percentiles. The threshold analysis repeats the complete fit at the 97.5th, 99th, and five-year citation means, the h_5 index, network centrality (Eigenfactor) (Bergstrom et al., 2008), prestige rank (SJR-like) (González-Pereira et al., 2010), mean article score, and context-normalised impact were computed. Journal 960 citation networks were constructed from references in the reference year (2024) to publications in the preceding two-, three-, or five-year windows, depending on the metric 99.5th percentiles for ten fixed seeds. A row with an unavailable required detector input is not zero-imputed; it receives no detector score, cannot be flagged, and is identified by a separate detector-completeness field. The overlap table is restricted to rows on which both criteria are defined.

965 C.1 Stage 3: Journal Subject Classification

Each journal was assigned to one of five broad subjects using an LLM-based classifier (see Appendix ??). The final mapping covers approximately 15 000 ISSNs. Within each subject, the 25th and 75th Eigenfactor percentiles were computed to define low-impact and high-impact thresholds, respectively The anomaly score is the negative of the library's

Table 12: Screen stability when one input is removed.

Detector feature omitted	Flags	Case flags	Control flags	Enrichment	Overlap with main screen
Self-citation rate	2,056	1,889 (2.49%)	167 (0.17%)	14.88	0.993
Coauthor-citation rate	1,901	1,753 (2.31%)	148 (0.15%)	15.58	0.931
Reciprocity	2,052	1,886 (2.48%)	166 (0.17%)	14.94	0.995
Local clustering	2,050	1,883 (2.48%)	167 (0.17%)	14.83	0.990
Outgoing HHI	1,907	1,758 (2.31%)	149 (0.15%)	15.52	0.934

Each fit uses the 99.0% Control percentile and seed 42. The main detector-complete population and cohesion restriction are held fixed; Case and Control percentages retain eligible-row denominators. Overlap is the Jaccard share of flagged rows retained by both screens.

`score_samples` output. The library’s `contamination=auto` decision cutoff is not used; the reported cutoff is computed from the subject-specific Control scores. 970

C.1 Stage 4: Author Population and Sampling

To keep computation tractable, the author population was sampled by retaining only ORCIDs whose last digit is 0 (approximately 10% of all ORCID-bearing authors in the Crossref snapshot). This deterministic filter preserves the distribution of author characteristics because the ORCID check digit is algorithmically assigned and uncorrelated with research behaviour.

C.1 Stage 5: Author Classification and Matching 975

Within the sampled population, authors were classified per subject (detailed in Section ??). Authors not meeting either criterion were excluded. A subject-specific

D Reproducibility Checks

The rebuild checks the following manuscript-facing invariants:

- (ORCID, s , tier) is unique in the feature output; 980
- generated pairs have complete subject-keyed h_5 index was computed for each remaining author. Case authors were then matched to Control authors via greedy one-pass matching within subject, as described in Appendix ?. records, satisfy the inclusive three-point caliper, and do not reuse an author–subject key;

- every analytic edge has non-null endpoints and is unique on (s, i, j, t) ;
- each per-metric complete-pair count is no greater than 102,626;
- every observed surge share lies in $[0, 1]$;
- the same final flag supplies every flag-dependent enrichment and figure artifact; and
- reported numerical values enter through generated LaTeX artifacts rather than hand-copied prose.

D.1 Stage 6: Final Dataset

The matching procedure yielded 9431 Case-Control pairs across five subjects (Table ??). For the maximum citation burst metric, 7711 pairs were available due to data sparsity. Directed author-to-author citation networks were constructed for

matched authors using fractional edge weights (Appendix ??). SQL fixtures separately test fractional weight, null endpoints, work-reference deduplication, subject retention, and prior-coauthor timing. Python unit tests cover full-ORCID cohort construction, matching eligibility and deterministic order, dynamic cohort validation, metric formulas, paired inference, unique clique membership, exact matched clique inference, detector overlap, feature ablation, weighted mixing, and flag reuse.

E Journal Subject Classification

F LLM-Based Journal Classification

Journal subject assignment used The retained journal mapping was produced with OpenAI gpt-4.1 (snapshot gpt-4.1-2025-04-14, accessed June 2025) with temperature 0 and deterministic seeding.

The prompt supplied the journal name and, where available, a representative abstract and article titles, together with few-shot exemplars for each category. Decision rules prioritised abstract content over article titles and journal names to

reduce ambiguity. Predictions were cached deterministically to a JSON file; the final mapping covers approximately 15 000 ISSN_s stored in the `issn_subjects` table of the study database. No independent ground-truth validation against human annotators was performed, gpt-4.1-2025-04-14) at zero temperature from journal names and representative work titles and abstracts. Prompts and cached outputs are a part of the replication materials. Because the mapping has no independent human-validation sample and no open-weight model was used as a baseline; both are acknowledged as limitations (Section 10) baseline, classification error remains a limitation rather than a measured source of uncertainty (LLM Guidelines Consortium, 2024).